# Machine Learning for Mechanical Engineers at CoEP Session 8: Linear Regression

## Session 8: LinRegr

# Linear Regression

### Want to know future? Predictions

- How does sales volume change with changes in price.
- How is this affected by changes in the weather?
- How does the amount of a drug absorbed vary with dosage and with body weight of patient?
- Does it depend on blood pressure?

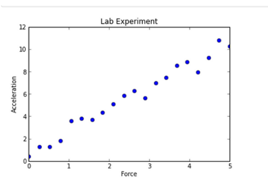
Answering questions like these, requires us to create a model.

### The model

- A model is a formula where one variable (response) varies depending on one or more independent variables (covariates).
- The formula (model) can be simple or complex.
- Simplest is a Linear Model where the dependent varies linearly with the independent variable(s).
- Creating a Linear Model involves a technique known as Linear Regression.

### Remember? - High School Physics Lab

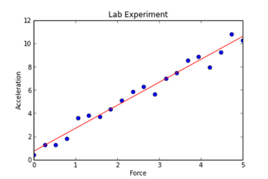
- Finding how acceleration changes when Force is applied
- For some input X (say force), got output Y (say acceleration).
- Plotted the pairs of observations x, y.




What next?

### Fitting line

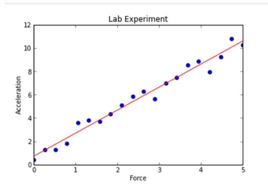
- Then you had to fit a straight line through points
- A visual "best fit", some points above, some below.
- Found'm' the slope (how?), and 'b' (how?).




Whats the model here?

### Fitting line

- Equation of line, the formula, itself is the model.
- You were doing informal Linear Regression.




### Linear Regression

- What: for predicting a quantitative variable
- Age: "it's been around for a long time". The term regression was devised by Francis Galton in his article Regression Towards Mediocrity in Hereditary Stature in 1886.
- Complexity: easy to understand and use
- Popularity: still widely used

### Possible Techniques

- Statisticians know Regression as OLS (Ordinary Least Square) method to fit a line. Deals with minimization of RMS (Root Mean Square) error.
- Theoretical/Analytical Solution $W = (X^T X)^{-1} X^T y$. Not practical for large matrices. Also, matrix is not invertible if any two rows are same.
- Machine Learning Engineers know Regression as prediction of continuous variables (floats). It uses Gradient Descent for the minimization part.

# Linear Regression - Analytical Approach (OLS)

### Example: Tip Amount

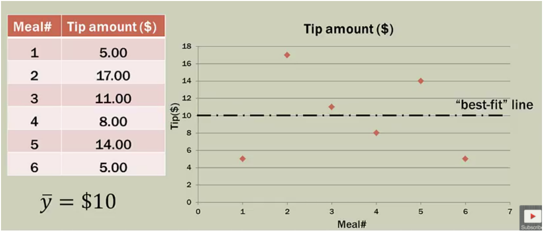
- In a restaurant, tip you pay to the waiter,
- Typically depends on the bill amount.
- Smaller the bill, lesser the tip, and vice versa.

Can we predict the tip amount just by looking at the bill amount?
Inputs?
Outputs?

### Example: Tip Amount

- You collect, random (meaning, any) 6 tips.
- YOU HAVE NOT COLLECTED THE BILL AMOUNT.
- Just one variable (Meal id is just the descriptor its not a independent variable)
- Any guesses?

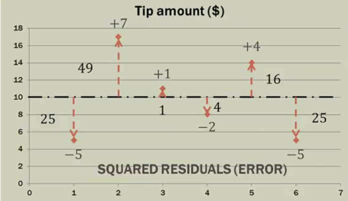
### Most Basic Situation




- With just one variable, mean is the best we can do.
- Note: not a single point falls on it.
- Its just an estimate.
- Goodness of fit?

(Reference: Simple Linear Regression: Step-By-Step - Dan Wellisch)

## Example: Tip Amount

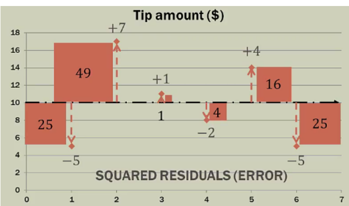



| Meal# | Residual | Residual² |
|---|---|---|
| 1 | −5 | 25 |
| 2 | +7 | 49 |
| 3 | +1 | 1 |
| 4 | −2 | 4 |
| 5 | +4 | 16 |
| 6 | −5 | 25 |

*Sum of squared errors (SSE) = 120*

- Std Variation says about goodness of fit.
- Diff of each point with mean are called 'residuals' or 'errors'
- Idea is to minimize SSE (Sum of Squared Errors) or RSS (Residual Sum of Squares)

## Important Concept



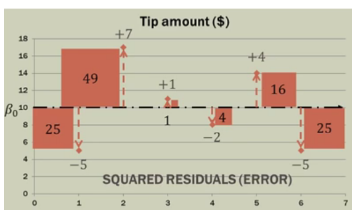

When conducting simple linear regression with TWO variables, we will determine how good that line "fits" the data by comparing it to THIS TYPE; where we pretend the second variable does not even exist.

If a two-variable regression model looks like this example, what does the other variable do to help explain the dependent variable?

NOTHING.

- With just the target variable (tips) we got SSE as 120.
- If we add one more variable to the problem, ie bill amount, what do you expect of SSE?
- It should reduce, or else, whats the use of adding this variable.
- So, we will compare our Linear regression fit line with this (one variable) line.

## Important Concept



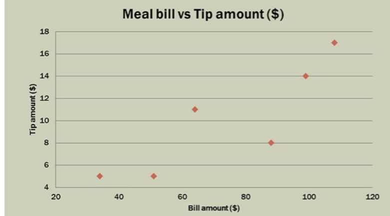

When conducting simple linear regression with TWO variables, we will determine how good the regression line "fits" the data by comparing it to THIS TYPE; where we pretend the second variable does not even exist; the slope, $\beta_1 = 0$.

In this situation, the value of $\hat{y}$ is 10 for every value of $x$.

$\hat{y} = b_0 + b_1 x$ $\quad b_0 = 10$
$\hat{y} = b_0 + (0)x$
$\hat{y} = b_0$ $\quad \hat{y} = 10$

- We find slope and intercept as $b_1$ and $b_0$ respectively.
- A line with $b_1 = 0$ is the horizontal line.
- Its SSE becomes the benchmark. Any selected pair of $b_1$ and $b_0$ should give SSE less than the benchmark SSE.

## Linear Regression with Two Variables

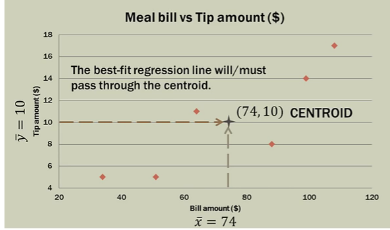



| Bill ($) | Tip ($) |
|---|---|
| 34.00 | 5.00 |
| 108.00 | 17.00 |
| 64.00 | 11.00 |
| 88.00 | 8.00 |
| 99.00 | 14.00 |
| 51.00 | 5.00 |

- You can fit a line, however you want.
- Even horizontal line can be put
- But then, which one is the best?
- Need to be at least better than horizontal line!!

## Methodology

Hypothesis: A line passing through the centroid of the data points will be good. Lets check.




| Bill ($) | Tip ($) |
|---|---|
| 34.00 | 5.00 |
| 108.00 | 17.00 |
| 64.00 | 11.00 |
| 88.00 | 8.00 |
| 99.00 | 14.00 |
| 51.00 | 5.00 |
| $\bar{x} = 74$ | $\bar{y} = 10$ |

- Calculate mean of both x and y.
- Line needs to pass through this point

## Calculations

$\hat{y}_i = b_0 + b_1 x_i$

Intercept

$$b_0 = \bar{y} - b_1 \bar{x}$$

Slope

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$\bar{x}$ = mean of the independent variable
$\bar{y}$ = mean of the dependent variable
$x_i$ = value of independent variable
$y_i$ = value of dependent variable

- Intercept and Slope are calculated as above.
- A simple tabular calculations can be done.

## Calculations

| Meal | Total bill ($) | Tip amount ($) | Bill deviation | Tip Deviations | Deviation Products | Bill Deviations Squared |
|---|---|---|---|---|---|---|
| | $x$ | $y$ | $x_i - \bar{x}$ | $y_i - \bar{y}$ | $(x_i - x)(y_i - \bar{y})$ | $(x_i - x)^2$ |
| 1 | 34 | 5 | -40 | -5 | 200 | 1600 |
| 2 | 108 | 17 | 34 | 7 | 238 | 1156 |
| 3 | 64 | 11 | -10 | 1 | -10 | 100 |
| 4 | 88 | 8 | 14 | -2 | -28 | 196 |
| 5 | 99 | 14 | 25 | 4 | 100 | 625 |
| 6 | 51 | 5 | -23 | -5 | 115 | 529 |
| | | | | | | |
| | $\bar{x} = 74$ | $\bar{y} = 10$ | | | $\sum = 615$ | $\sum = 4206$ |

- Slope comes to $= 615/4206 = 0.1462$
- Intercept comes to $= 10 - 0.1462(74) = -0.818$
- So, eq is $y = 0.1462x - 0.8188$

## Interpretation

$$\hat{y}_i = 0.1462x - 0.8188$$

For every \$1 the bill amount ($x$) increases, we would expect the tip amount to increase by \$0.1462 or about 15-cents.

If the bill amount ($x$) is zero, then the expected/predicted tip amount is \$-0.8188 or negative 82-cents! Does this make sense? NO. The intercept may or may not make sense in the "real world."

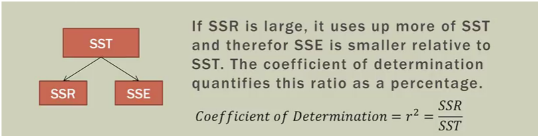
- Slope gives proportionality with which tip is calculated.
- Intercept may or may not make sense in real life.
- Is this manual calculation giving good line?

## Evaluation

| Meal | Total bill ($) | Observed tip amount ($) | $\hat{y}_i$ (predicted tip amount) | Error $(y - \hat{y}_i)$ | Squared Error $(y - \hat{y}_i)^2$ |
|---|---|---|---|---|---|
| | $x$ | $y$ | | | |
| 1 | 34 | 5 | 4.1505 | 0.8495 | 0.7217 |
| 2 | 108 | 17 | 14.9693 | 2.0307 | 4.1237 |
| 3 | 64 | 11 | 8.5365 | 2.4635 | 6.0688 |
| 4 | 88 | 8 | 12.0453 | −4.0453 | 16.3645 |
| 5 | 99 | 14 | 13.6535 | 0.3465 | 0.1201 |
| 6 | 51 | 5 | 6.6359 | −1.6359 | 2.6762 |
| | | | | | |
| | $\bar{x} = 74$ | $\bar{y} = 10$ | | | SSE= $\sum = 30.075$ |

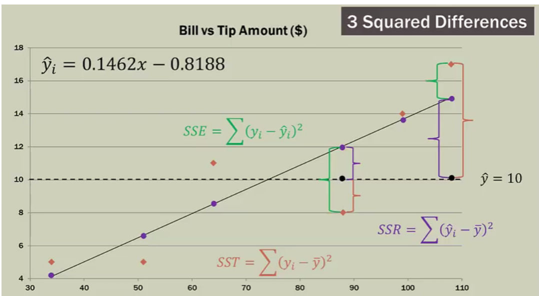
- Sum of Squares of Error , SSE is 30 for our manual model
- Its far less than the horizontal line benchmark of 120.
- So, by adding Bill amount as independent variable, reduced error by about 90.

## Evaluation




- SST (sum of squares total) is the benchmark error. Its constant ie 120.
- SSE (sum of squares error) is the level to which we could get error to it by applying regression ie 30.
- The difference, the achievement by regression is called SSR (sum of squares by regression) ie 90.

## Evaluation




- For each x, there is predicted y and actual y
- 3 relationships: observed, predicted and total
- Can decide quality based on the ratios among them.

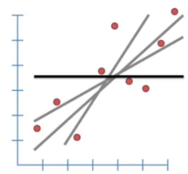
## $R^2$ Statistics

- $SST = SSR + SSE$
- $SST$: Total variance in the response Y. Amount of variability inherent in the response, before the regression is performed. $y_i$ to $\bar{y}$
- It has two components, one explained by this regression model, other the unexplained.
- $SSR$: The amount of variability that is explained after performing the regression. $\hat{y}_i$ to $\bar{y}$
- $SSE$ : The amount of variability that is left unexplained, that our model could not minimize. $y_i$to $\hat{y}_i$

# Linear Regression - Machine Learning Approach

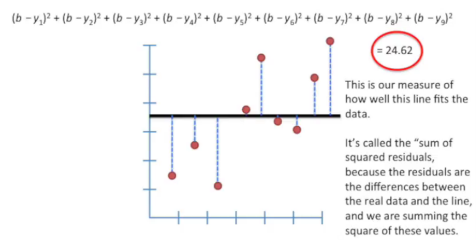
## Linear regression

- We have multiple observations of inputs and an output
- Object: to find a model, ie a formulation, adhering to more or less all data points
- ie. to attempt to find the "best" line
- But, which line is the "best"?
- The horizontal line is the worst, the benchmark. We need to beat that!!



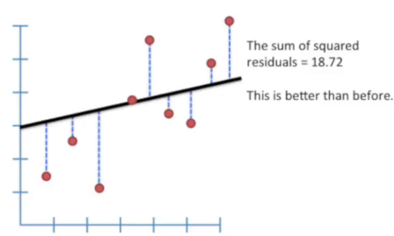
(Ref: StatQuest: Linear Models - Josh Starmer )

## Sum of Squared Errors for Benchmark Line




(Ref: StatQuest: Linear Models - Josh Starmer )

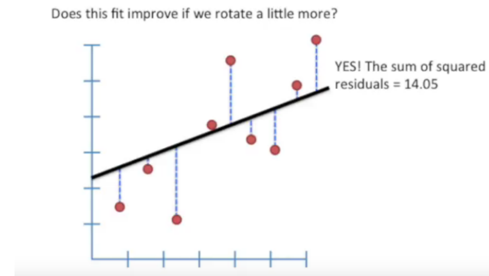
## Sum of Squared Errors for Rotated Line




Its better but is this best?
(Ref: StatQuest: Linear Models - Josh Starmer )

## Sum of Squared Errors for Rotated Line

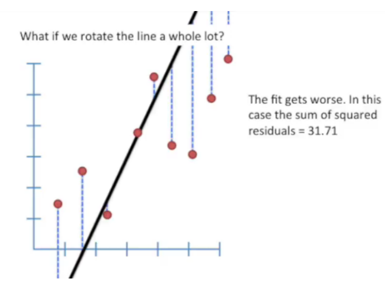
Rotated a bit more. Better?




Its better but is this best?
(Ref: StatQuest: Linear Models - Josh Starmer )

## Sum of Squared Errors for Rotated Line

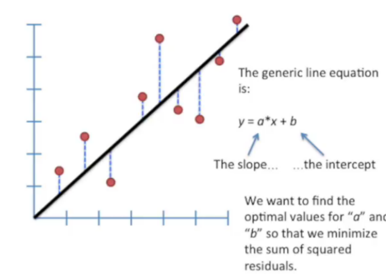
Why not rotate a whole lot!!




No, we lost the BEST point somewhere in between. Whats that "optimized" point?
(Ref: StatQuest: Linear Models - Josh Starmer )

## Optimization

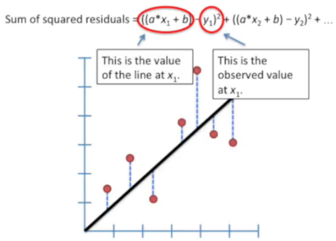
To find the sweet spot ie the optimized point, lets put some formulation based on generic line equation.




(Ref: StatQuest: Linear Models - Josh Starmer )

## Optimization

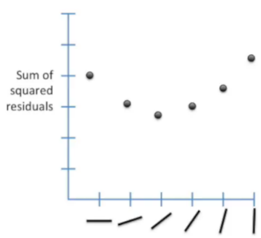
Mathematically . . .




Since we want the line that will give us the smallest sum of the squares of errors (SSE: Sum of Squared Errors), this method for finding the best values for "a" and "b" is called as "Least Squares".
(Ref: StatQuest: Linear Models - Josh Starmer )

## Optimization

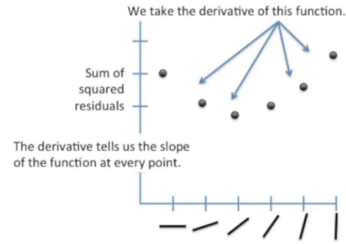
- If we plotted SSE for each rotated line the plot looks like.
- We can see that SSE was high for the horizontal line, then came down as we started rotating the line,
- But started rising again, if we passed the sweet spot.




(Ref: StatQuest: Linear Models - Josh Starmer )

## Optimization

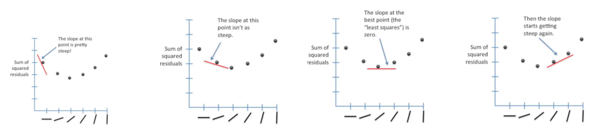
- We need to get to the bottom point.
- We can start at any (random) point, and SLIDE down to bottom.
- SLIDING means going along slope.
- Slope is given by Derivative of the (SSE) function/curve, at the point




(Ref: StatQuest: Linear Models - Josh Starmer )

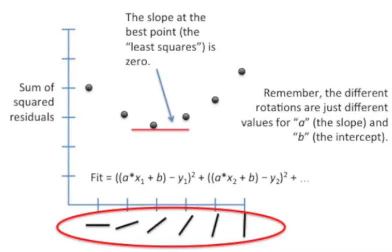
## Optimization

- Slope starts decreasing from end, faster,then slower,
- Reaches 0, horizontal, and the starts increasing again.
- Need to find the "0" slope point.



These different slopes means different "a" parameters (Ref: StatQuest: Linear Models - Josh Starmer )

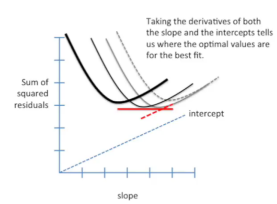
## Optimization




(Ref: StatQuest: Linear Models - Josh Starmer )

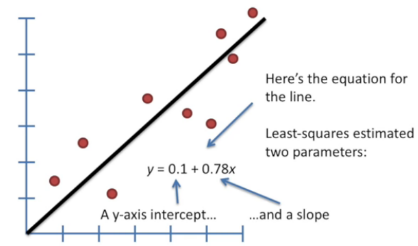
## Optimization

- For "a" and "b" both to be optimized, we will need 3D picture, as the Loss function would then take Z axis
- It will be a Surface.
- Derivative will be Partial, but the objective is the same, Bottom-most point. ie Minimum Loss function value




The '"a" and "b" coordinates corresponding to the bottom-most point give most minimum Loss. So it represents the "best" line.
(Ref: StatQuest: Linear Models - Josh Starmer )

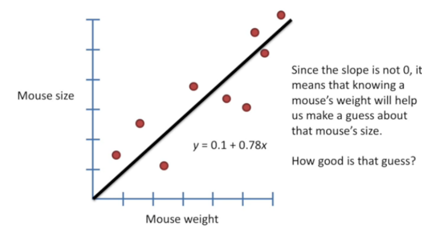
## Results




(Ref: StatQuest: Linear Models - Josh Starmer )

## Example: Mouse Size

- We have data points having mouse size and mouse weight
- We wish to find mouse size given mouse weight.




we use $R^2$ to find "good-ness" of fit (of line)
(Ref: StatQuest: Linear Models - Josh Starmer )

## Linear regression For Complex Problems

- In real life, problems are far more complex that just one x and one y.
- There could be many x's ie many inputs which decide the output
- General formulation : Predicting quantitative response $y$ based on predictor variables $x$. Assumes linear relationship between $x$ and $y$

$$y_i = x_{1,i}\beta_1 + x_{2,i}\beta_2 + \cdots + x_{p,1}\beta_p + \epsilon_i$$

- Objective is to find the parameters $\beta_j$ that minimize the squared residuals
- There is no slope formulas for these $\beta$s. So, we will solve machine learning way.

# Linear Regression - Advertising Budget Example

## Example: Advertising Dataset

- Toy Dataset: Advertising, Advertising.csv
- Sales totals for a product in 200 different markets. Advertising budget in each market, broken down into TV, radio, newspaper

| | TV | Radio | Newspaper | Sales |
|---|---|---|---|---|
| **0** | 230.1 | 37.8 | 69.2 | 22.1 |
| **1** | 44.5 | 39.3 | 45.1 | 10.4 |
| **2** | 17.2 | 45.9 | 69.3 | 9.3 |
| **3** | 151.5 | 41.3 | 58.5 | 18.5 |
| **4** | 180.8 | 10.8 | 58.4 | 12.9 |

(Data: http://www-bcf.usc.edu/g̃areth/ISL/data.html)

## Problem

- Goal: What marketing plan for next year will result in high product sales?
- Questions (Hypothesis):
  - Is there a relationship between advertising budgets and sales?
  - How strong is the relationship between advertising budgets and sales?
  - Which media contribute to the sales the most?
  - Is the relationship linear?

## EDA

Correlation Matrix

| | TV | Radio | Newspaper | Sales |
|---|---|---|---|---|
| **TV** | 1.000000 | 0.054809 | 0.056648 | 0.782224 |
| **Radio** | 0.054809 | 1.000000 | 0.354104 | 0.576223 |
| **Newspaper** | 0.056648 | 0.354104 | 1.000000 | 0.228299 |
| **Sales** | 0.782224 | 0.576223 | 0.228299 | 1.000000 |

- Correlation between radio and newspaper is 0.35
- Barely any correlation (or "not correlated") for TV/radio and TV/newspaper

## EDA

| | TV | Radio | Newspaper | Sales |
|---|---|---|---|---|
| **TV** | 1.000000 | 0.054809 | 0.056648 | 0.782224 |
| **Radio** | 0.054809 | 1.000000 | 0.354104 | 0.576223 |
| **Newspaper** | 0.056648 | 0.354104 | 1.000000 | 0.228299 |
| **Sales** | 0.782224 | 0.576223 | 0.228299 | 1.000000 |

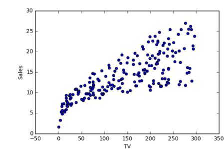
- Reveals tendency to spend more on Newspaper advertising in markets where more is spent on Radio advertising.
- Sales higher in markets where more is spent on Radio, but more also tends to be spent on Newspaper.
- In Simple LM: Newspaper "gets credit" for effect of Radio on Sales.

## Advertising Dataset

- What are some ways we can regress sales onto adverting using Simple Linear Regression?
- One model:

$$\begin{aligned} sales &\approx \beta_0 + \beta_1 \times TV \\ y &\approx \beta_0 + \beta_1 \times x \end{aligned}$$

- Scatter plot visualization for TV and Sales.

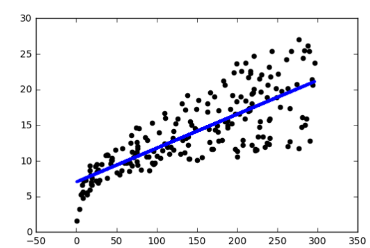


## Advertising Dataset

- Simple Linear Model in Python (using pandas and scikit): Predictor: $x$, Response: $y$
- $Sales = 7.03259 + 0.04754 \times TV$

## Advertising Dataset

Scatter plot visualization for TV and Sales with Linear Model.




If we have 3 such models, can we "somehow" compose them (say, averaging) to get the combined model?

## Multiple Linear Regression

- In practice, often have more than one predictor
  - $sales \approx \beta_{0,1} + \beta_{1,1} \times TV$
  - $sales \approx \beta_{0,2} + \beta_{1,2} \times Newspaper$
  - $sales \approx \beta_{0,3} + \beta_{1,3} \times radio$
- Run three separate simple linear regressions
- For p predictor variables: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p + \epsilon$
- Since error $\epsilon$ has mean zero, we usually omit it.
- A one-unit change in any predictor variable $x_i$ will change the expected mean response by $\beta_j$ units. $sales = \beta_0 + \beta_1 \times TV + \beta_2 \times radio + \beta_3 \times newspaper$

## Estimating the Parameters $\beta_0, \beta_1, \ldots$

- Advertising Dataset $sales = \beta_0 + \beta_1 \times TV + \beta_2 \times radio + \beta_3 \times newspaper$
- After solving $sales = 2.93 + 0.0457 \times TV + 0.188 \times radio + -0.0010 \times newspaper$

## Coefficients

- Many Simple Linear Regression
  - $TVModel$ : [[0.04753664]][7.03259355]
  - $RadioModel$ : [[0.20249578]][9.3116381]
  - $NewspaperModel$ : [[0.0546931]][12.35140707]
- Single Multiple Linear Regression
  - Coefficients: `[[0.04576465, 0.18853002, -0.00103749]]`
  - $Intercept$ : [2.93888937]

## Answering questions using Regression Model

Goal: What marketing plan for next year will result in high product sales?

- Question: Is there a relationship between advertising budget and sales?
- Answer: Yes, hypothesis testing shows that we can reject the null hypothesis that $B_{TV} = B_{RADIO} = B_{NEWSPAPER} = 0$

## In conclusion

- Pros of Linear Regression Model:
  - Provides nice interpret-able results
  - Works well on many real-world problems
- Cons of Linear Regression Model:
  - Assumes linear relationship between response and predictors: Change in the response Y due to a one-unit change in Xi is constant
  - Assumes additive relationship. Effect of changes in a predictor Xi on response Y is independent of the values of the other predictors